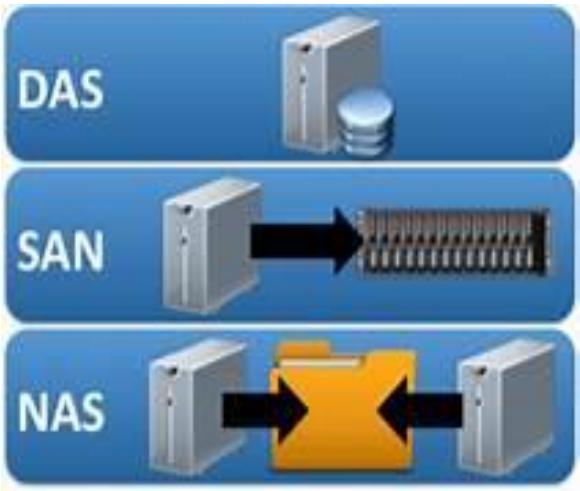


CLOUD COMPUTING COURSE

Lecture19- Cloud Storage (2)

ABSTRACT

Lecture19 Introduces centralized storage types in networks.

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Centralized Storage

Centralized storage means that all storage resources are centrally deployed and are provisioned over a unified interface.

With centralized storage, all physical disks are centrally deployed in disk enclosures and are used to provide storage services externally through the controller. Centralized storage typically refers to disk arrays.

A disk array combines multiple physical disks into a single logical unit. Each disk array consists of one controller enclosure and multiple disk enclosures. This architecture delivers an intelligent storage space featuring **high availability**, **high performance**, and **large capacity.**

Disk Array→



Based on the technical architecture, centralized storage can be categorized as:

DAS, NAS and SAN.

1. DAS (Direct attached storage):

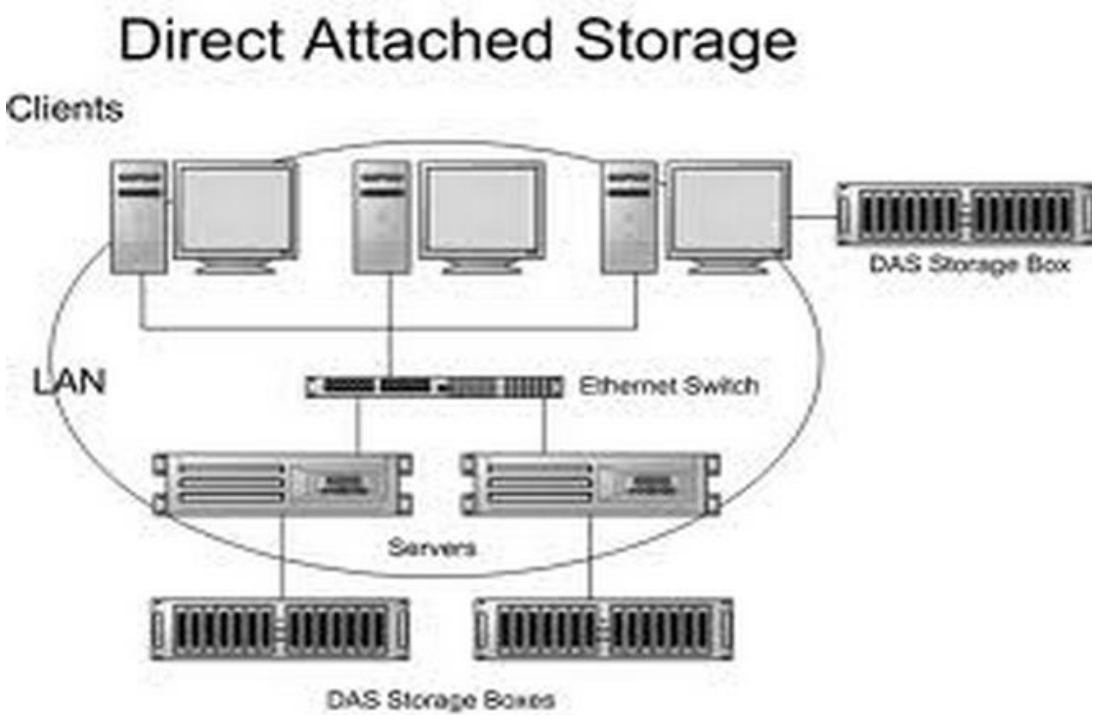
It is digital storage directly attached to the computer accessing it.

Examples of DAS include hard drives, solid-state drives, optical disc drives, and storage on external drives. DAS is the most widely used storage system before SAN is introduced.

DAS Types:

- Standalone
- JBOD (Spanned)
- RAID





2. NAS (Network attached storage)

A technology integrates distributed and independent data into a large-scale and centralized data center for access by different hosts and application servers.

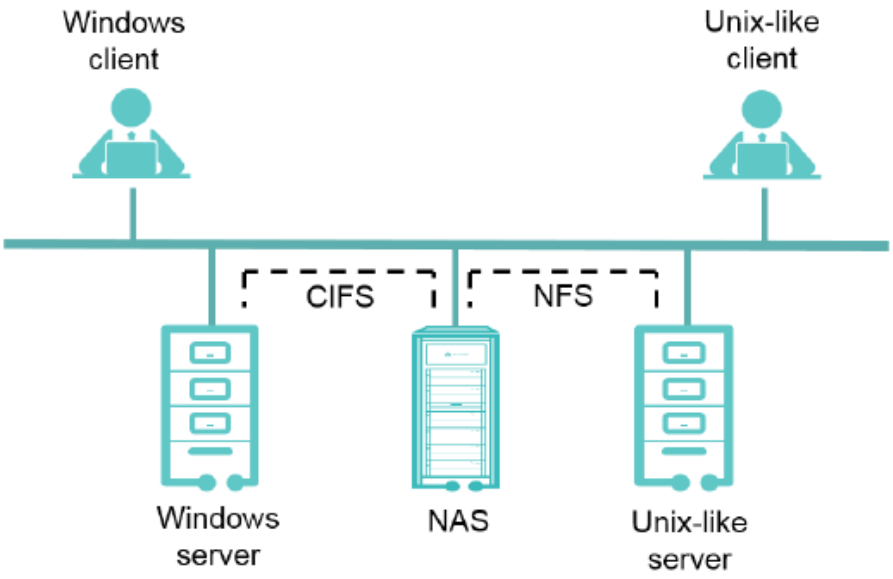
- NAS is **a file-level computer** data storage connected to a computer network providing data access to heterogeneous group of clients.
- An NAS server **contains storage devices**, such as disk arrays, CD/DVD drives, tape drives, or portable storage media.
- An NAS server delivers an embedded **OS to share files** across platforms.

Simply speaking, a NAS is a device that is connected to the network with file storage function. Therefore, NAS is also called a **network file storage device.**

- It is often manufactured as a computer appliance – a purpose-built specialized computer
- It delivers centralized file storage and management and separates storage devices from servers, **releasing bandwidth and improving performance.**

Essentially, NAS is a storage device rather than a server.

- The advantage of NAS is that it can deliver file storage services in a **fast and cost-effective** manner using existing resources in the data center. The current solution is compatible between UNIX, Linux, and Windows OSs, and can be easily connected to users' TCP/IP networks. The following shows the NAS system.



CIFS vs NFS are protocols developed to permit a client system to access the file system on remote computing devices like server or personal computers **(File Sharing Protocols).**

1. **NFS** is the "**Network File System**" specifically used for **Unix and Linux operating systems** systems to share files and folders.. It allows files communication transparently between servers and end users machines like desktops & laptops. NFS uses client- server methodology to allow user to view read and write files on a computer system.
2. **CIFS** is abbreviation for "**Common Internet File System**" used by Windows operating systems for file sharing. CIFS also uses the client-server methodology where a client makes a request of a server program for accessing a file .The server takes the requested action and returns a response. CIFS is an open standard version of the Server Message Block Protocol (SMB) developed and used by Microsoft and it uses the **TCP/IP protocol.**

